

Personal Info

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angeloskatsampekis@gmail.com

Nationality

Greek

Date of birth

19/11/1994

Links

[LinkedIn](#)

Skills

Unity 3D / C# (HDRP, URP, XR Interaction)

VR/AR development (Meta Quest, Valve Index, OpenXR, Meta XR SDKs)

Game Development

Multiplayer networking (Netcode for GameObjects)

Multithreaded physics & gameplay systems

ML-Agents & Generative-AI pipelines

ONNX, NNAPI, TensorFlowLite, Sentsis

Cesium 3D Tiles & Geospatial data

C++, Python tooling, Git, CI/CD

Mobile Optimization: GLES, Vulkan, Android NDK

Data Visualization

Angelos-Ioannis Katsampekis

Unity developer and research engineer

I'm a **Unity Developer** and **XR Research Engineer** with 5+ years of experience creating games and immersive experiences. I've worked on projects for **Samsung, Coca-Cola, and EU Horizon programs**, ranging from VR avatar-tracking systems and AI-powered museums to a full drone-and-missile simulator with advanced physics and ML-based navigation. My background includes strong experience with **Unity, C#, C++, and Python**, as well as practical work in gameplay programming, physics simulation, multiplayer networking, machine learning, generative AI, and performance optimization.

Work Experience

Research Assistant, CERTH - Center for Research & Technology, Hellas, Thessaloniki

2022 - 2025

Samsung Electronics - Avatar Body Capture (Unity)

- **Prototyped** a VR full-body avatar estimation system in Unity, trained on the **AMASS dataset** (SMPL format) to reconstruct body joint rotations from only head and hand inputs (position, rotation, and velocities).
- **Integrated** multiple inference back-ends (**Sentis, ONNX Runtime, TensorFlow Lite**) to enable real-time performance for VR/XR devices.
- **Built** an in-app **benchmarking framework** to evaluate model variants and inference engines under identical runtime conditions, optimizing the pipeline for VR use cases.

Drone & Missile Simulator (Unity)

- **Designed** and built the entire **simulator** end-to-end, from core architecture to UX.
- **Developed** and optimized a custom real-time, multithreaded **aerodynamic solver** with per-vertex force computation for accurate flight simulation.
- **Trained** autonomous drones with **Unity ML-Agents** to navigate dense forest environments using a depth camera.
- **Logged** missile-launch telemetry generating labeled datasets for **AI targeting research**.

Palimpsisto - 3D Geospatial Platform (Unity)

- **Developed** an interactive **geospatial platform** for the Archaeological site of Kythnos, funded by the Greek government.

- **Integrated Cesium** to stream photogrammetry scans as 3D tiles from a Geospatial Database.
- **Connected** an **AI-Assistant** for in-platform interaction.
- **Linked** live museum findings **metadata** allowing curators to update exhibits.

ReEvaluate - AI-Driven VR Museum (EU Horizon, Unity)

- **Developed** an immersive museum where **generative AI** creates themed textures and auto-places artifacts based on semantic descriptions.

5G VR Automotive Tour (Unity Multiplayer)

- **Developed** a **VR Multiplayer** app in Unity that receives a 360-degree video stream over 5G, applies foveated rendering and face anonymization on the fly and reports networking metrics.

VirTourArt Platform (EU Horizon, Unity SDK)

- **Delivered** AR/VR rendering & multimodal interaction modules, localization & 3D reconstruction and packaged the subsystems into an **SDK**.

EMBNOESIS - Intelligent NB-IoT sensors for the construction industry

- **Developed** a Python **data-visualization tool** that converts live NB-IoT sensor streams into interactive dashboards for construction-site monitoring.

Coca-Cola - CCH: Software Development Service

- **Designed** and implemented machine learning **algorithms** for Coca-Cola, working alongside their data science team to support inventory optimization research.

Software Developer, Pragma - IoT Solutions, Thessaloniki

2024 - 2025

Samsung Electronics - Game Optimization

- **Created** Python and C++ **tools** for Samsung to convert and package ONNX models to ANN for mobile AI runtime, streamlining deployment on Android devices.
- **Benchmarked** mobile **games** with a custom graphics monitoring harness, capturing frame-time, thermal, and battery data to guide optimization in collaboration with Samsung's engineering team.

Education

Physics, Aristotle University of Thessaloniki (BSc)

2020

**Advanced Computer and Communication Systems, Aristotle
University of Thessaloniki (MSc)**

2025

**Data Science with Python, National and Kapodistrian University of
Athens (Certificate)**

2021